MOHAMMED ZAKWAN M

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TECHNICAL SKILLS

Core Programming: Python (Proficient), C, Java, SQL, JavaScript, HTML/CSS (Fundamentals)

Computer Vision & AI/ML: Object Detection (YOLO Architectures v1–v26), From-Scratch Model Implementation, Image Processing, Deep Learning Pipelines, Neural Networks

Frameworks & Libraries: PyTorch, OpenCV, NumPy, Pandas, Matplotlib

Developer Tools: Git, GitHub, VS Code, Google Colab, Jupyter Notebook, Antigravity

Core CS Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems

EDUCATION

Bachelor of Technology in Computer Engineering (Artificial Intelligence & Machine Learning) 2023 – 2027

Presidency University, Bangalore, Karnataka | CGPA: 7.24 / 10.0

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering, Storage Architecture

PROJECTS

Aerial & UAV Object Detection & Tracking System

Python, PyTorch, YOLO, OpenCV, TensorRT, ONNX | github.com/zakwanmunawar/AeroDrop-Tracker

- Trained a 2.37M-parameter YOLO26n object detection model on a 12-class aerial benchmark dataset (86,000+ annotations), achieving 22.1% mAP@50 across all target classes under high background clutter and object density.
- Optimized real-time execution pipelines on an NVIDIA Tesla T4 GPU, achieving sub-3ms total frame latency (330+ FPS raw inference speed).
- Applied dataset preprocessing, frame resizing, and bounding-box normalization to improve detection accuracy across varying background conditions.
- Integrated OpenCV for real-time video feed inference with dynamic bounding-box tracking and minimal latency.

Construction Worker Safety & PPE Compliance Detector

Python, YOLO, OpenCV | github.com/zakwanmunawar/worker-safety-system

- Developed and fine-tuned a 3M-parameter YOLO object detection pipeline for workplace site safety, achieving 97.0% mAP@50 and 96.5% precision across 3 critical violation classes (Fall, Helmet, Jacket).
- Optimized video feed inference pipelines on an NVIDIA Tesla T4 GPU, processing multi-class site compliance alerts at sub-9ms frame latency (117+ FPS end-to-end execution).
- Implemented OpenCV video feed processing to overlay dynamic, color-coded bounding boxes flagging missing equipment for automated site monitoring.

CERTIFICATIONS & ACHIEVEMENTS

YOLO Architecture Implementation & Research Series (v1 to v26)

PyTorch & Google Colab

- Implemented YOLOv1 from scratch in PyTorch; systematically analyzing and implementing object detection model progressions across YOLO iterations.

Virtual Internship Certification in Python Programming – CodSoft 2025

- Successfully completed a 4-week virtual program developing modular Python scripts, core algorithmic logic, and OOP concepts.

Data Structures & Algorithms in Python – NeetCode 150

- Solved the NeetCode 150 problem set in Python, covering all core patterns — arrays & hashing, two pointers, sliding window, trees, graphs, dynamic programming, and backtracking.

Top 70 Selection (out of 500+ Teams) – University Innovation & Project Review (IPR)

- Recognized for hardware-software integration on the Autonomous Robotic Car Arm project.